

```
[14]: score = 91

if score >= 85:
    print("Grade : A")
elif score >= 70:
    print ("Grade : B")
elif score >= 60:
    print ("Grade: C")
elif score >= 50:
    print ("Grade: D")
else:
    print ("Grade: F")
```

Grade : A

```
[15]: score = 76
attendance = 92

if score >= 50 and attendance >= 75:
    print ("Validated")
elif score >= 50 and attendance < 75:
    print ("Blocked: Low attendance")
else:
    print ("Failed")
```

Validated

```
[16]: #Question 1.3
#What would go wrong if elif score >= 50: came before elif score >= 70:? Answer in one sen
# condition would never be reached for scores 70 and above.
```

```
[19]: scores = [72, 85, 91, 64, 78, 88, 55, 93, 61]

for score in scores:
    if score >= 80:
        print("Top score:", score)
```

Top score: 85
Top score: 91
Top score: 88
Top score: 93

```
[25]: scores = [72, 85, 91, 64, 78, 47, 88, 55, 93, 61]
```

```
total = 0
count = 0
passed = 0
failed = 0

for score in scores:
    total = total + score
    count = count + 1

    if score >= 50 :
        passed = passed + 1
    else:
        failed = failed + 1

average = total / count
print ("total:", total)
print ("count:", count)
print ("average:", average)
print ("passed:", passed, "\nFailed:", failed)
```

```
total: 734
count: 10
average: 73.4
passed: 9
Failed: 1
```

```
[30]: goal = 250000
monthly = 15000
saved = 0
months = 0

while saved < goal:
    saved = saved + monthly
    months = months + 1

print ("Months needed:", months)
```

```
Months needed: 17
```

[32]: `def get_grade(score):`

```
    if score >= 85:
        return "A"
    elif score >= 70:
        return "B"
    elif score >= 60:
        return "C"
    elif score >= 50:
        return "D"
    else:
        return "F"
print(get_grade(91))
print(get_grade(76))
print(get_grade(43))
```

A
B
F

[34]: `def pass_rate(score_list):`

```
    passed = 0
    count = 0
    for s in score_list:
        count = count + 1
        if s >= 50 :
            passed = passed + 1
    return passed * 100 / count

print (pass_rate (scores))
print (pass_rate ([80, 45, 30, 90]))
```

90.0
50.0

[39]: `with open("week3_students.csv") as f:`
 `lines = f.readlines()`

```
print("Number of lines:", len(lines))
print("Header:", lines[0].strip())
print("First student:", lines[1].strip())
```

Number of lines: 41
Header: student_id,name,age,gender,program,district,attendance_pct,score
First student: AUCA001,Aline Uwase,19,F,Information Systems,Gasabo,92,85

```
[40]: names = []
      for line in lines[1:]:
          parts = line.strip().split(",")
          names.append(parts[1])

      print("Count:", len(names))
      print("First 5:", names[:5])
```

Count: 40
First 5: ['Aline Uwase', 'Eric Niyonzima', 'Diane Mukamana', 'Jean Bosco Habimana', 'Sandrine Ingabire']

```
[42]: valid = 0
      bad = 0
      bad_ids = []
      total = 0

      for line in lines[1:]:
          parts = line.strip().split(",")
          try:
              score = int(parts[7])          # risky line: may raise ValueError
              total = total + score          # only reached if int() worked
              valid = valid + 1
          except ValueError:
              bad = bad + 1
              bad_ids.append(parts[0])      # remember which student_id was broken

      average = total / valid

      print("Valid:", valid)
      print("Bad:", bad)
      print("Bad IDs:", bad_ids)
      print("Average of valid scores:", round(average, 1))
```

Valid: 37
Bad: 3
Bad IDs: ['AUCA008', 'AUCA020', 'AUCA032']
Average of valid scores: 75.3

```
[43]: passed = 0
failed = 0
best_score = -1
best_name = ""

for line in lines[1:]:
    parts = line.strip().split(",")
    try:
        score = int(parts[7])
        if score >= 50:
            passed = passed + 1
        else:
            failed = failed + 1
        if score > best_score:
            best_score = score
            best_name = parts[1]
    except ValueError:
        pass # broken row: ignore it (valid rows only)

print("Passed:", passed)
print("Failed:", failed)
print("Top student:", best_name, "with", best_score)
print("Top student's grade:", get_grade(best_score))

Passed: 35
Failed: 2
Top student: Bonaventure Nkurunziza with 95
Top student's grade: A
```

```
[45]: sums = {}
sums["Gasabo"] = 85
print(sums.get("Gasabo", 0))
print(sums.get("Huye", 0))

85
0
```

```
[46]: sums = {}          # district -> running total of scores
      counts = {}       # district -> how many valid scores

      for line in lines[1:]:
          parts = line.strip().split(",")
          try:
              s = int(parts[7])
              d = parts[5]
              sums[d] = sums.get(d, 0) + s
              counts[d] = counts.get(d, 0) + 1
          except ValueError:
              pass          # broken row: skip it

      best_district = ""
      best_avg = -1

      for d in sums:
          avg = sums[d] / counts[d]
          print(d, "->", round(avg, 1), "| students:", counts[d])
          if avg > best_avg:
              best_avg = avg
              best_district = d

      print("Highest average:", best_district, round(best_avg, 1))
```

```
Gasabo -> 80.4 | students: 5
Musanze -> 65.2 | students: 5
Muhanga -> 87.0 | students: 5
Kicukiro -> 64.0 | students: 4
Huye -> 68.4 | students: 5
Nyagatare -> 82.8 | students: 5
Nyarugenge -> 74.8 | students: 5
Rubavu -> 78.7 | students: 3
Highest average: Muhanga 87.0
```

```
[47]: with open("week3_report.txt", "w") as f:
      f.write("Week 3 Report\n")
      f.write("Students in file: " + str(len(lines) - 1) + "\n")
      f.write("Valid scores: " + str(valid) + "\n")
      f.write("Bad scores: " + str(bad) + "\n")
      f.write("Average of valid scores: " + str(round(average, 1)) + "\n")
      f.write("Top student: " + best_name + " (" + str(best_score) + ")\n")

      with open("week3_report.txt") as f:
          print(f.read())
```

```
Week 3 Report
Students in file: 40
Valid scores: 37
Bad scores: 3
Average of valid scores: 75.3
Top student: Bonaventure Nkurunziza (95)
```